

Do Vision-Language Models See Visualizations Like Humans?

Alignment in Chart Categorization

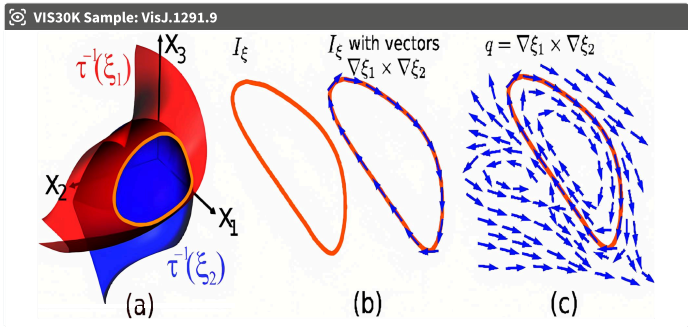


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One Image, Two Realities

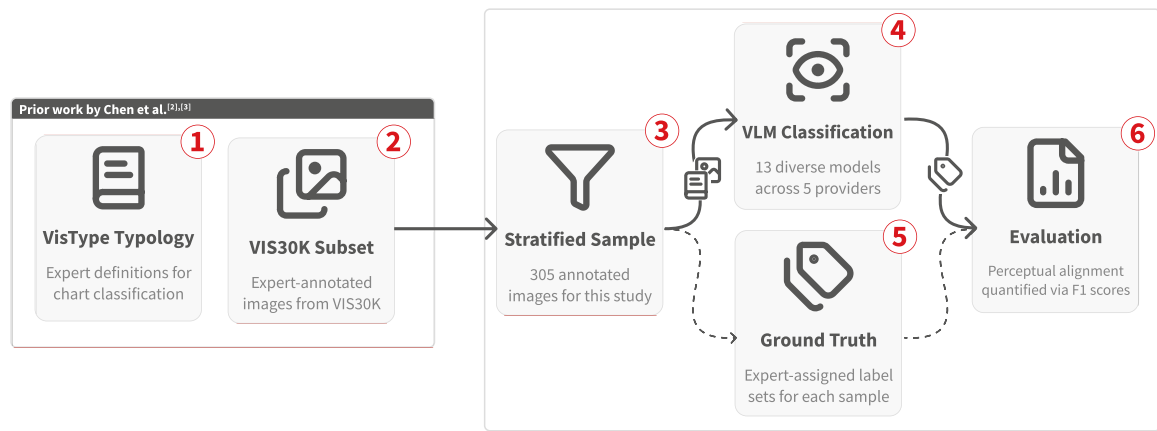
Human Expert Perception	
Purpose	✓ visualization
Encoding	✓ line ✓ surface-volume ✓ glyph
Dimensionality	✓ 3D



VLM Perception of GPT-4.1	
Purpose	✗ schematic
Encoding	✓ line ✓ surface-volume ✗ point
Dimensionality	✗ 2D ✓ 3D

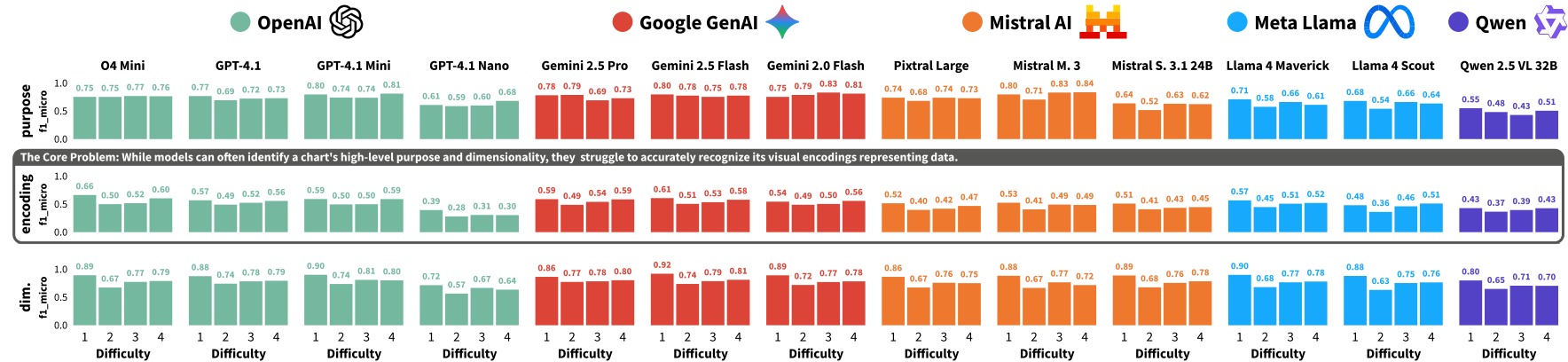
To become a useful partner, an AI must first perceive charts like a human. Here, that foundational skill fails at every level: the figure's **purpose** is misidentified, a core **encoding** is mistaken, and a **dimension** is hallucinated. To build reliable AI, we must first understand this gap, making systematic measurement essential.

Benchmarking Perceptual Alignment



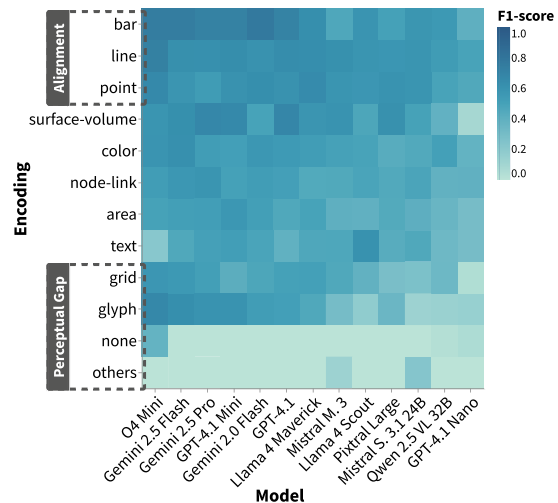
- Guiding Framework:** The VisType typology^[21] provided a shared, objective classification framework for both human experts and VLMs.
- Expert-Annotated Dataset:** A foundational dataset of 6,000+ images from VIS30K^[43], annotated by human experts according to the VisType typology^[21].
- Stratified Sample:** A 305-image stratified sample was drawn from the expert dataset to ensure a balanced and representative test set.
- Perceptual Test:** In a strict zero-shot task, 13 diverse VLMs classified each image, guided only by the raw visual input and the VisType^[21] definitions provided as their system prompt.
- Benchmark:** The expert labels for the 305-image sample—specifying each image's *purpose*, *encoding*, and *dimensionality*—served as the ground truth for evaluating human-VLM alignment.
- Alignment Measurement:** Perceptual alignment was quantified by comparing VLM predictions against the ground truth using the multi-label F1-Score.

The Anatomy of Misalignment



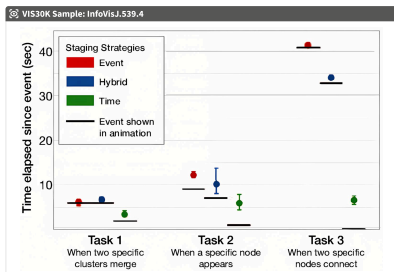
Encoding Performance Breakdown

This map of F1 scores reveals a clear performance divide: while alignment is strong for simple encodings, a significant perceptual gap exists for ambiguous types and when human experts abstained from labelling.



A Failure of Consensus

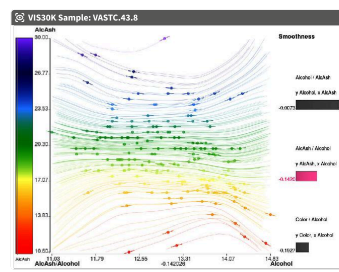
Flagship vision-language models not only fail on the same complex image, but they also disagree on the incorrect alternative, revealing a lack of shared perceptual understanding between VLMs.



Perceived Encoding	
Human Expert	✓ bar ✓ point ✗ line ✗ point ✗ line ✗ text ✗ line ✗ color
Gemini 2.5 Pro	✓ point
Pixtral Large	✓ point
Qwen 2.5 VL 32B	✓ point

A Challenge Beyond Scale

This counter-intuitive trend, where smaller models can outperform larger ones on the encoding task, suggests the perceptual gap is a deep architectural challenge, not merely a problem of model scale.



Perceived Encoding	
Human Expert	✓ color ✓ line ✗ color ✓ line ✗ point ✗ bar ✗ point
Llama 4 Scout	✓ line
Llama 4 Maverick	✓ line
Gemini 2.5 Pro	✗ point

References

- [1] A. Bendeck and J. Stasko. An Empirical Evaluation of the GPT-4 Multimodal Language Model on Visualization Literacy Tasks. *IEEE Transactions on Visualization and Computer Graphics*, 31(1):1105–1115, 2025. doi:10.1109/TVCG.2024.3456155
- [2] J. Chen, P. Isenber, R. S. Laramée, T. Isenber, M. Sedimair, T. Möller, R. S. Laramée, H.-W. Shen, K. Wünsche, and Q. Wang. VIS30K: A Collection of Figures and Tables from IEEE Visualization Conference Publications. *IEEE Transactions on Visualization and Computer Graphics*, 27(9):3826–3833, 2021. doi:10.1109/TVCG.2021.3054916
- [3] J. Chen, M. Ling, R. Li, P. Isenber, T. Isenber, M. Sedimair, T. Möller, R. S. Laramée, H.-W. Shen, K. Wünsche, and Q. Wang. VIS30K: A Collection of Figures and Tables from IEEE Visualization Conference Publications. *IEEE Transactions on Visualization and Computer Graphics*, 27(9):3826–3833, 2021. doi:10.1109/TVCG.2021.3054916
- [4] J. Hong, C. Seto, A. Fan, and R. Maciejewski. Do LLMs have visualization literacy? an evaluation on modified visualizations to test generalization in data interpretation. *IEEE Transactions on Visualization and Computer Graphics*, pp. 1–13, 2025. doi:10.1109/TVCG.2025.3536358

